

Practice by Standard: Number Sense

Standard Set 1.0: Students understand the place value of whole numbers and decimals to two decimal places and how whole numbers and decimals relate to simple fractions. Students use the concepts of negative numbers.

DIRECTIONS

Choose the best answer.

QUICK Practice

- 1** Which of these is the number 4,008,020? **(4NS1.1)**
- A** four million, eight thousand, twenty
 - B** four million, eighty thousand, twenty
 - C** four thousand, eight thousand, twenty
 - D** four billion, eight thousand, twenty

- 2** Which of the following has the greatest value? **(4NS1.2)**
- F** 21.3 **H** 7.19
G 18.5 **J** 6.32

- 3** What is 614,782 rounded to the nearest hundred? **(4NS1.3)**
- A** 600,000
 - B** 610,000
 - C** 614,800
 - D** 615,000

QUICK Review

STRATEGY Use the place values of the digits in the number to read it.

How would you read the number 4,008,020?

For more help with reading whole numbers, see page 22.

STRATEGY Line the decimal points up to compare the numbers.

Which decimal has the greatest tens digit?

For more help with comparing decimals, see page 566.

STRATEGY If the digit in the tens place is 5 or more, round up. If it is 4 or less, round down.

What is the digit in the tens place?
Should you round up or down to the nearest hundred?

For more help with rounding whole numbers, see page 36.

QUICK Practice

- 4 Which fraction represents the smallest part of a whole? (4NS1.5)

F $\frac{1}{10}$

G $\frac{1}{8}$

H $\frac{1}{5}$

J $\frac{1}{4}$

- 5 Which fraction means the same as 0.23? (4NS1.6)

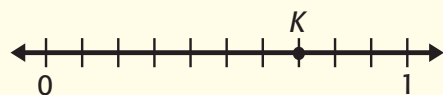
A $\frac{23}{1}$

B $\frac{23}{10}$

C $\frac{23}{100}$

D $\frac{23}{1000}$

- 6 What fraction is best represented by point K on this number line? (4NS1.9)



F $\frac{3}{4}$

G $\frac{7}{10}$

H $\frac{2}{3}$

J $\frac{2}{5}$

QUICK Review

STRATEGY Look at the denominators to compare the fractions.

Which fraction has the largest denominator?

For more help with comparing fractions, see page 528.

STRATEGY How would you read the decimal 0.23?

What are the place values of the digits 2 and 3 in 0.23? How do you read 0.23?

For more help with understanding fraction and decimal equivalents, see page 572.

STRATEGY Count the number of dashes between 0 and 1. What does each dash represent?

What fractional part of a whole does each dash mark represent?

For more help with identifying fractions on a number line, see page 528.

Practice on Your Own

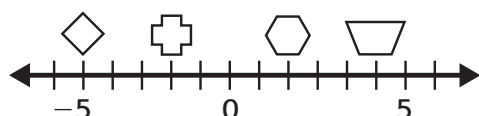
Standard Set 1.0

- 7** The numbers below increase by the same amount each time. What are the next three numbers in this pattern? (4NS1.8)

7, 10, 13, 16, 19, —, —, —

- A** 21, 24, 27 **C** 22, 24, 27
B 22, 25, 28 **D** 23, 26, 29

- 8** Which symbol is located at -2 on the number line below? (4NS1.8)



- 9** Which of the following has the greatest value? (4NS1.2)
- A** thirty thousand, eight hundred, fifty-nine
B thirty thousand, five hundred, eighty-nine
C thirty thousand, fifty-eight
D thirty thousand, five hundred, eighty-five

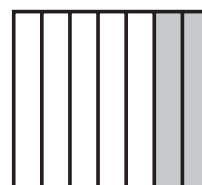
- 10** The population of a certain country is about eighty-three million people. What is this in standard form? (4NS1.1)

- F** 83000
G 8,300,000
H 80,300,000
J 83,000,000

- 11** What is 45,715,903 rounded to the nearest thousand? (4NS1.3)

- A** 45,715,000
B 45,715,900
C 45,716,000
D 45,720,000

- 12** What fraction is represented by the shaded part of the figure below? (4NS1.7)



- F** $\frac{1}{7}$ **H** $\frac{2}{7}$
G $\frac{1}{4}$ **J** $\frac{3}{8}$

- 13** Which of the following means the same as 0.17? (4NS1.6)
- A** seven hundredths
B seventeen hundredths
C seventeen thousandths
D seven thousandths

Practice by Standard: Number Sense

Standard Set 2.0: Students extend their use and understanding of whole numbers to the addition and subtraction of simple decimals.

DIRECTIONS

Choose the best answer.

QUICK Practice

- 1** On Monday, Belinda practiced the piano for 42 minutes. On Tuesday, she practiced for 29 minutes, and on Wednesday she practiced for 47 minutes. About how many minutes did she spend practicing the piano all three days? (4NS2.1)

A 100 minutes **C** 150 minutes
B 120 minutes **D** 200 minutes

- 2** Raul had \$15.84 in his wallet. He spent \$4.79 on a package of light bulbs. About how much did he have left over after buying the light bulbs? Round your answer to the nearest dollar. (4NS2.1)

F \$9 **H** \$11
G \$10 **J** \$12

- 3** The total length of an insect is 1.28 centimeters. What is the length of the insect rounded to the nearest tenth? (4NS2.2)

A 1 cm **C** 1.22 cm
B 1.2 cm **D** 1.3 cm

QUICK Review

STRATEGY Use rounding to estimate the total number of minutes.

Round each amount of minutes to the nearest ten. Then find the sum.

For more help with estimating the sum of whole numbers, see page 58.

READING HINT The words *left over* mean subtraction. Use rounding to estimate the difference between the two amounts.

Round each amount up to the nearest whole dollar. Then subtract the amounts to estimate how much he has left over.

For more help with estimating the difference of decimals, see page 598.

STRATEGY Look at the digit in the hundredths place to know how to round.

Should you round up or down when the next digit is 8?

For more help with rounding decimals, see page 593.

QUICK Practice

- 4 What is the height of the tree rounded to the nearest whole number? (4NS2.2)



- F 200 inches
- G 210 inches
- H 215 inches
- J 220 inches

QUICK Review

STRATEGY Look to the digit one place to the right of the ones digit. If it is 5 or greater, round the ones digit up. If it is 4 or less, round down.

What is the tenths digit in 215.34?

For more help with rounding decimals, see page 593.

Hundreds	Tens	Ones	Tenths	Hundredths
2	1	5	3	4

- 5 Meredith wants to purchase a stapler, an ink cartridge, and a calculator for her home office.

Office Supplies	
Stapler	\$9.89
Ink Cartridge	\$19.95
Calculator	\$28.05

About how much will she spend altogether for the supplies? Round your answer to the nearest whole number. (4NS2.1)

- A \$58
- B \$60
- C \$65
- D \$67

STRATEGY Use rounding to estimate the total cost of the supplies.

Round the cost of each item to the nearest whole number. Then find the sum.

What is the closest ten to each number?

For more help with estimating the money, see page 598.

Practice on Your Own

Standard Set 2.0

- 6** What is the length of the skateboard rounded to the nearest whole number?

(4NS2.2)

- F** 22 inches **H** 22.6 inches
G 22.5 inches **J** 23 inches

- 7** Levi drives a van for a delivery company. The table shows how many miles he drove each day last week.

Day	Miles
Monday	195
Tuesday	247
Wednesday	408
Thursday	353
Friday	296

About how many miles did Levi drive altogether last week? (4NS2.1)

- A** 1000 miles **C** 1500 miles
B 1200 miles **D** 1800 miles

- 8** Taka's cat has a tail with a length of 11.76 inches. What is the length of her cat's tail rounded to the nearest tenth?

(4NS2.2)

- F** 12 inches **H** 11.7 inches
G 11.8 inches **J** 11 inches

- 9** Mrs. Sato spent \$3.89 on apples, \$8.09 on juice, and \$2.99 on cereal at the grocery store. About how much did she spend altogether? Round your answer to the nearest dollar. (4NS2.1)

- A** \$15 **C** \$17
B \$16 **D** \$18

- 10** The total length of an airplane is 38.65 feet. What is the length of the airplane rounded to the nearest whole number? (4NS2.2)

- F** 38 feet **H** 38.7 feet
G 38.6 feet **J** 39 feet

- 11** Thomas had \$35.21 in his piggy bank. He earned \$14.75 for mowing his neighbor's lawn and put this in the bank. About how much did he have in the piggy bank altogether? Round your answer to the nearest dollar. (4NS2.1)

- A** \$45 **C** \$55
B \$50 **D** \$60

- 12** Jamal bought a sandwich for \$4.79 and a juice for \$1.15. He paid with \$20. About how much change did he receive? Round your answer to the nearest dollar. (4NS2.1)

- F** \$12 **H** \$14
G \$13 **J** \$15

Practice by Standard: Number Sense

Standard Set 3.0: Students solve problems involving addition, subtraction, multiplication, and division of whole numbers and understand the relationships among the operations.

DIRECTIONS

Choose the best answer.

QUICK Practice

1 $4189 - 2785 =$ (4NS3.1)

- A** 2414
- B** 2404
- C** 1414
- D** 1404

2 $493 \div 7 =$ (4NS3.2)

- F** 7 R3
- G** 70 R3
- H** 71
- J** 71 R3

- 3** There are 12 teams of 14 students at the school field day. How many students are participating in field day altogether? (4NS3.3)

- A** 168
- B** 154
- C** 48
- D** 26

QUICK Review

STRATEGY Line the numbers up in a vertical column and find the difference.

Use borrowing from digits with larger place values if needed.

For more help with subtracting whole numbers, see page 70.

STRATEGY First decide how many times 7 goes into 49. Then continue to use the method for dividing whole numbers.

Use the method for dividing whole numbers to find the quotient.

For more help with dividing whole numbers, see page 358.

READING HINT The word *of* usually indicates a multiplication problem.

You can solve the problem by finding the product of 12 and 14.

For more help with solving problems that involve multiplication, see page 310.

QUICK Practice

- 4 Lakota rode 52 miles on her bicycle in 4 hours. She rode at the same speed during these 4 hours. What was her average speed in miles per hour? (4NS3.4)

F 8 miles per hour
G 10 miles per hour
H 12 miles per hour
J 13 miles per hour

- 5
$$\begin{array}{r} 3751 \\ + 4160 \\ \hline \end{array}$$
 (4NS3.1)

A 5413
B 6421
C 7911
D 8011

- 6 There are 7 school buses at Horizon Elementary School this morning. Each bus holds the same number of students. If there are 266 students on the buses altogether, how many students are on each bus? (4NS3.4)

F 36
G 38
H 178
J 259

QUICK Review

STRATEGY Use division to find Lakota's average speed.

This problem can be solved by finding the quotient of 52 and 4.

For more help with solving problems that involve division of whole numbers, see page 352.

STRATEGY Use rounding to estimate the sum. Then eliminate answers that seem unreasonable.

Add the digits in each column from right to left. If the sum in a column is greater than 9, add a 1 to the next column.

For more help with adding whole numbers, see page 64.

STRATEGY Use division to find the number of students on each bus.

You can solve this problem by finding the quotient of 266 and 7.

For more help with solving problems that involve division of whole numbers, see page 358.

Practice on Your Own

Standard Set 3.0

- 7** Simona solved the problem below. Which expression could be used to check her answer? **(4NS3.2)**

$$654 \div 25 = 26 \text{ R}4$$

- A** $(25 \times 26) \times 4$
B $(25 \times 26) + 4$
C $(25 + 26) + 4$
D $(25 + 26) \times 4$

- 8** There are 200 booths at a state fair. Each booth is given 5000 tickets to sell to customers. How many tickets is this in all? **(4NS3.3)**

- F** 10,000
G 100,000
H 1,000,000
J 10,000,000

- 9**
$$\begin{array}{r} 8543 \\ - 3614 \\ \hline \end{array}$$
 (4NS3.1)

- A** 4919 **C** 5919
B 4929 **D** 5929

- 10** $842 \div 26 =$ **(4NS3.2)**

- F** 31 **H** 32
G 31 R9 **J** 32 R10

- 11** There are 15 mail carriers in Meera's hometown. Each mail carrier begins the day with 12 totes of letters to deliver. How many totes of letters are there in all? **(4NS3.3)**

- A** 27 **C** 180
B 37 **D** 210

- 12** Rudy's heart beat 510 times in 6 minutes. His heart rate was the same during this time period. How many times per minute did Julian's heart beat? **(4NS3.4)**

- F** 85 **H** 80
G 83 **J** 78

- 13** There are 9 bookshelves. Each shelf holds the same number of books. If there are 234 books on the shelves altogether, how many books are on each shelf? **(4NS3.4)**

- A** 24
B 26 **D** 28

- 14** A pet store had nine aquarium tanks of 36 fish each. Which expression can be used to find the total number of fish? **(4NS3.3)**

- F** $36 + 9$
G $36 - 9$
H 36×9
J $36 \div 9$

Practice by Standard: Number Sense

Standard Set 4.0: Students know how to factor small whole numbers.

DIRECTIONS

Choose the best answer.

QUICK Practice

- 1** Which of these is another way to write the product 8×7 ? **(4NS4.1)**
- A** $4 \times 3 \times 7$ **C** $8 \times 3 \times 4$
B $3 \times 2 \times 7$ **D** $4 \times 2 \times 7$

- 2** The ages of Christine and her siblings are shown in the table below.

Sibling	Age (years)
Christine	10
Jody	8
Kristen	15
Shawn	13
Robbie	12

Who has an age that is a prime number? **(4NS4.2)**

- F** Robbie **H** Kristen
G Shawn **J** Jody

- 3** Which is a composite number? **(4NS4.2)**
- A** 6 **C** 11
B 7 **D** 13

QUICK Review

STRATEGY Rewrite 8 as a product of two factors.

Can the number 7 be factored? Can the number 8?

For more help with factoring, see page 204.

READING HINT A *prime number* is a number that does not have any factors other than 1 and itself.

Which number only has factors of 1 and itself?

For more help with identifying prime numbers, see page 208.

READING HINT A *composite number* is a number that has more than two factors.

Which number has more than two factors?

For more help with identifying composite numbers, see page xxx.

QUICK Practice

4 Which statement is true? (4NS4.2)

- F** The only factors of 14 are 1 and 14.
- G** The only factors of 12 are 1 and 12.
- H** The only factors of 11 are 1 and 11.
- J** The only factors of 10 are 1 and 10.

5 Which of the following shows all of the factors of 18? (4NS4.1)

- A** 2, 3, 6, 9
- B** 2, 3, 6, 9, 18
- C** 1, 2, 3, 6, 7, 9, 18
- D** 1, 2, 3, 6, 9, 18

6 Which is a prime number? (4NS4.2)

- F** 6 **H** 8
- G** 7 **J** 9

7 Which of these is another way to write 12? (4NS4.1)

- A** 2×4 **C** $2 \times 2 \times 3$
- B** $2 \times 2 \times 2$ **D** $2 \times 3 \times 3$

QUICK Review

STRATEGY Choose the answer choice that shows a prime number.

Which of the numbers in the answer choices is prime?

For more help with identifying prime numbers, see page 208.

READING HINT A number is a *factor* of another number if it divides evenly into the other number.

How many ways can you multiply two numbers to get 18?

For more help with factoring numbers, see page 204.

READING HINT A *prime number* is a number that does not have any factors other than 1 and itself.

Find the number that only has 1 and itself as factors.

For more help with identifying prime numbers, see page xxx.

STRATEGY Write the number 12 as the product of its factors.

Multiply the numbers in the answer choices to see which one equals 12.

For more help with factoring numbers, see page 204.

Practice on Your Own

Standard Set 4.0

- 8** Which is a prime number? (4NS4.2)

F 14
G 15
H 16
J 17

- 9** Which statement is true? (4NS4.2)

A The only factors of 12 are 1 and 12.
B The only factors of 13 are 1 and 13.
C The only factors of 14 are 1 and 14.
D The only factors of 15 are 1 and 15.

- 10** Which of these is another way to write the product 15×11 ? (4NS4.1)

F $3 \times 5 \times 11$
G $2 \times 3 \times 5 \times 11$
H $3 \times 4 \times 5 \times 7$
J $2 \times 6 \times 11$

- 11** Erina practiced her clarinet every day for 20 minutes each day. Which of the following is another way to write 20? (4NS4.1)

A $2 \times 2 \times 5$
B $2 \times 3 \times 4$
C $2 \times 3 \times 5$
D $2 \times 4 \times 5$

- 12** The numbers of students in 4 different homerooms are shown.

Homeroom Teacher	Number of Students
Mr. Perez	25
Miss Foster	27
Mrs. Simpson	29
Mr. Ramos	28

Which homeroom teacher has a prime number of students in his or her homeroom? (4NS4.2)

F Mr. Perez
G Miss Foster
H Mrs. Simpson
J Mr. Ramos

- 13** Which of these is another way to write 30? (4NS4.1)

A $2 \times 2 \times 5$
B $2 \times 3 \times 4$
C $2 \times 3 \times 5$
D $2 \times 4 \times 5$

- 14** Suppose Dario starts counting from 2 through 16 as shown. How many prime numbers will he count? (4NS4.2)

2, 3, 4, 5, 6, ...

F 7 **H** 5
G 6 **J** 4

Practice by Standard: Algebra and Functions

Standard Set 1.0: Students use and interpret variables, mathematical symbols, and properties to write and simplify expressions and sentences.

DIRECTIONS

Choose the best answer.

QUICK Practice

- 1** Which number is represented by d in the number sentence below?

(4AF1.1)

$$7 \times d = 105$$

- A** 14 **C** 16
B 15 **D** 17

QUICK Review

STRATEGY Eliminate incorrect answer choices by substituting them into the number sentence.

Which answer choice results in a true number sentence when you substitute it for d ?

For more help with understanding variables in an equation, see page 236.

- 2** What is the value of the expression below? (4AF1.2)

$$(4 \times 7) \div (3 + 1)$$

- F** 5 **H** 7
G 6 **J** 8

STRATEGY Simplify within each set of parentheses first. Then perform the division.

Use the order of operations to simplify the expression.

For more help with evaluating numerical expressions, see page 230.

- 3** $4 \times (3 - 1)$ (4AF1.3)

- A** 11 **C** 9
B 10 **D** 8

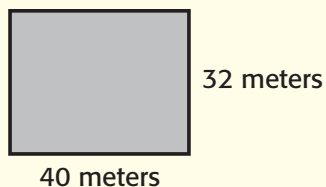
STRATEGY Remember to simplify within the parentheses first.

Use the order of operations to simplify the expression.

For more help with evaluating numerical expressions, see page 230.

QUICK Practice

- 4 Which equation below represents the area (A) of the rectangle in square meters? (4AF1.4)



- F** $A = 40 + 32$
G $A = 40 \times 32$
H $40 = A \times 32$
J $A = (2 \times 40) + (2 \times 32)$

- 5 The sum of a and b equals 20. If $a = 11$, which equation can be used to find b ? (4AF1.5)

- A** $11 + b = 20$
B $11 - b = 20$
C $b - 11 = 20$
D $b + 20 = 11$

- 6 $(6 \times 7) \div (7 - 4)$ (4AF1.2)

- F** 9
G 10
H 12
J 14

QUICK Review

STRATEGY The *area* of a rectangle can be found by multiplying the length by the width.

What is the length of the rectangle?
What is the width? How can you use these values to find the area?

For more help with using formulas, see page 444.

STRATEGY Represent the situation with an equation.

Write an expression for the sum of a and b and set it equal to 20. Then substitute 11 for a .

For more help with setting up equations to find unknown values, see page 98.

STRATEGY Simplify within each set of parentheses first. Then perform the division.

Use the order of operations to simplify the expression.

For more help with evaluating numerical expressions, see page 230.

Practice on Your Own

Standard Set 1.0

- 7** What is the value of the expression below if $m = 6$? (4AF1.2)

$$23 - (m + 9)$$

- A** 26 **C** 8
B 18 **D** 4

- 8** Look at the problem below.

$$\square + 19 = \triangle$$

If $\square = 11$, what is $\triangle$? (4AF1.5)

- F** 30
G 22
H 18
J 8

- 9** What is the value of b ? (4AF1.2)

$$(18 \div 3) \times (6 - 3) = b$$

- A** 2
B 12
C 14
D 18

- 10** $12 \div (6 - 2) =$ (4AF1.3)

- F** 0 **H** 3
G 2 **J** 4

- 11** What is the value of k ? (4AF1.2)

$$(15 \div 3) \times (6 + 4) = k$$

- A** 50 **C** 15
B 34 **D** 5

- 12** The formula $F = 32 + (C \times 9) \div 5$ takes a temperature in degrees Celsius (C) and tells you the temperature in degrees Fahrenheit (F). Suppose a thermometer reads 20°C . What is the temperature in degrees Fahrenheit? (4AF1.4)

- F** 62°F
G 68°F
H 70°F
J 73°F

- 13** $10 \times (7 - 3) + 5 =$ (4AF1.3)

- A** 36
B 45
C 60
D 72

- 14** $(11 \times 5) \div (4 + 1) =$ (4AF1.2)

- F** 7
G 9
H 11
J 15

- 15** The product of h and k equals 84. If $h = 12$, which equation can be used to find the value of k ? (4AF1.5)

- A** $12 \div k = 84$
B $84 - k = 12$
C $12 \times 84 = k$
D $12 \times k = 84$

Practice by Standard: Algebra and Functions

Standard Set 2.0: Students know how to manipulate equations.

DIRECTIONS

Choose the best answer.

QUICK Practice

- 1** The letters R and H stand for numbers. If $R + 65 = H + 65$, which statement is true? (4AF2.1)

A $R > H$ **C** $R = H$
B $R < H$ **D** $R = H - 65$

- 2** What number goes in the box to make this number sentence true? (4AF2.2)

$$(8 - 2) \times 3 = \square \times 3$$

F 5 **H** 7
G 6 **J** 8

- 3** $89 + \square = 89 + 12$ (4AF2.1)

A 2×2 **C** 2×4
B 2×3 **D** 3×4

QUICK Review

STRATEGY Recognize that if you add 65 to two equal numbers, the results are equal.

What do you know about the two numbers R and H that must be true if $R + 65 = H + 65$?

For more help with understanding that equals added to equals are equal, see page 110.

STRATEGY Recognize that if you multiply the same number by 3, the results are equal.

You can solve this problem by simplifying $8 - 2$.

For more help with understanding that equals multiplied by equals are equal, see page 248.

STRATEGY Eliminate choices first that do not meet all of the requirements.

Substitute each answer choice into the box to see which one results in a true number sentence.

For more help with understanding that equals added to equals are equal, see page 110.

QUICK Practice

- 4 What number goes in the box to make this number sentence true?

(4AF2.2)

$$\square \div 5 = (22 + 8) \div 5$$

- F 30
- G 18
- H 10
- J 6

QUICK Review

STRATEGY Recognize that if you divide the same number by 5, the results are equal.

You can solve this problem by simplifying $22 + 8$.

For more help with understanding that equals divided by equals are equal, see page 248.

- 5 $9 \times \square = 9 \times 7$ (4AF2.2)

- A 3
- B 7
- C 49
- D 63

STRATEGY Eliminate choices first that do not meet all of the requirements.

Substitute each answer choice into the box to see which one results in a true number sentence.

For more help with understanding that equals multiplied by equals are equal, see page 248.

- 6 The letters C and D stand for numbers. If $C - 32 = D - 32$, which statement is true? (4AF2.1)

- F $C = D$
- G $C < D$
- H $C = D + 32$
- J $C = D - 32$

STRATEGY Recognize that if you subtract 32 from two equal numbers, the results are equal.

What do you know about the two numbers C and D that must be true if $C - 32 = D - 32$?

For more help with understanding that equals subtracted from equals are equal, see page 110.

Practice on Your Own

Standard Set 2.0

- 7** The letters M and N stand for numbers. If $M \times 6 = N \times 6$, which statement is true? (4AF2.2)

A $M > N$
B $M = N$
C $M < N$
D $M = N \div 6$

- 8** What number goes in the box to make this number sentence true? (4AF2.1)

$$40 + 7 = \square + 7$$

F 7 **H** 40
G 33 **J** 47

- 9** What number goes in the box to make this number sentence true? (4AF2.1)

$$\square - 9 = (3 \times 5) - 9$$

A 15 **C** 11
B 13 **D** 6

- 10** What number goes in the box to make this number sentence true? (4AF2.2)

$$81 \div \square = 81 \div 3$$

F 3 **H** 18
G 9 **J** 27

- 11** What expression goes in the box to make this number sentence true? (4AF2.1)

$$17 + 16 = 17 + \square$$

A 2×4 **C** 4×4
B 3×4 **D** 3×8

- 12** What number goes in the box to make this number sentence true? (4AF2.2)

$$\square \div 4 = (9 + 7) \div 4$$

F 4
G 8
H 12
J 16

- 13** What number goes in the box to make this number sentence true? (4AF2.2)

$$5 \times (7 - 4) = 5 \times \square$$

A 3
B 5
C 7
D 15

- 14** The letters Q and R stand for numbers. If $Q + 56 = R + 56$, which statement is true? (4AF2.1)

F $Q = R + 56$
G $Q = R - 56$
H $Q = R$
J $Q < R$

- 15** What expression goes in the box to make this number sentence true? (4AF2.2)

$$\square \times 9 = 15 \times 9$$

A 2×3
B 2×5
C 3×4
D 3×5

Practice by Standard: Measurement and Geometry

Standard Set 1.0: Students understand perimeter and area.

DIRECTIONS

Choose the best answer.

QUICK Practice

- 1 Which statement about the figures is true? (4MG1.2)

Figure 1

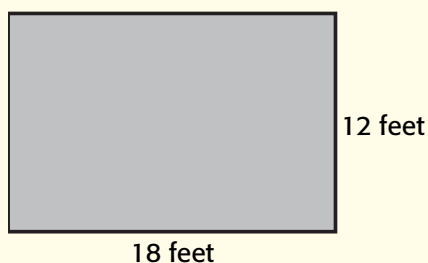
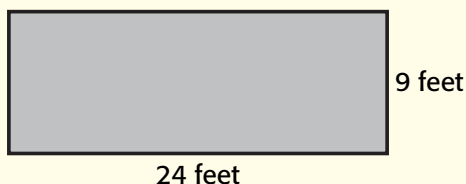


Figure 2



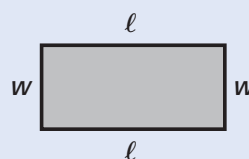
- A** They both have the same width.
- B** They both have the same length.
- C** They both have the same area.
- D** They both have the same perimeter.

QUICK Review

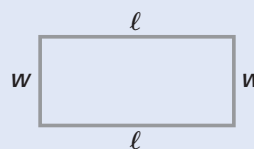
STRATEGY Compare the areas and perimeters of the two rectangles.

What is the area of each rectangle?
What is the perimeter of each rectangle?

Recall that area is the number of square units needed to cover a figure.



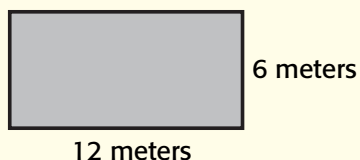
Perimeter is the distance around a figure.



For more help with recognizing that rectangles that have the same area can have different perimeters, see page 448.

QUICK Practice

- 2 What is the area of the rectangle below? (4MG1.1)



- F 80 square meters
- G 72 square meters
- H 42 square meters
- J 36 square meters

- 3 Which statement about the figures is true? (4MG1.3)

Figure 1

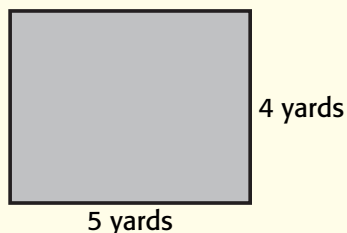
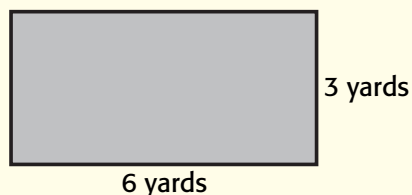


Figure 2



- A They both have the same width.
- B They both have the same length.
- C They both have the same area.
- D They both have the same perimeter.

QUICK Review

STRATEGY Use the formula for the area of a rectangle, $A = l \times w$.

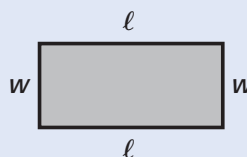
Find the area of the rectangle by multiplying the length by the width.

For more help with finding the area of a rectangle, see page 444.

STRATEGY Compare the areas and perimeters of the two rectangles.

What is the area of each rectangle?
What is the perimeter of each rectangle?

Recall the formulas for the area and perimeter of a rectangle.



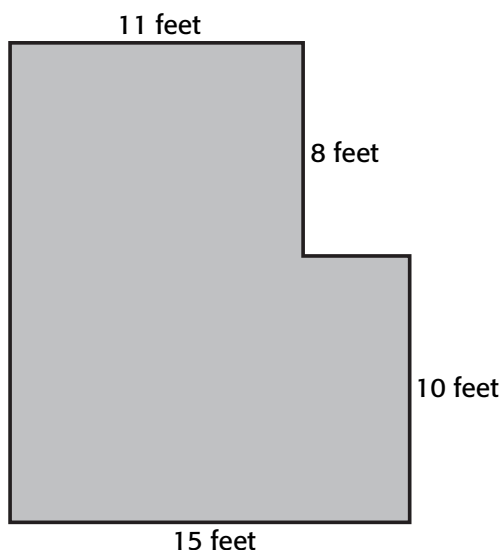
$$A = l \times w \quad P = 2l + 2w.$$

For more help with recognizing that rectangles that have the same perimeter can have different areas, see page 448.

Practice on Your Own

Standard Set 1.0

- 4** Mr. Romero's deck has the shape and dimensions shown below.



What is the area of the deck? (4MG1.4)

- F** 270 square feet
- G** 256 square feet
- H** 238 square feet
- J** 220 square feet

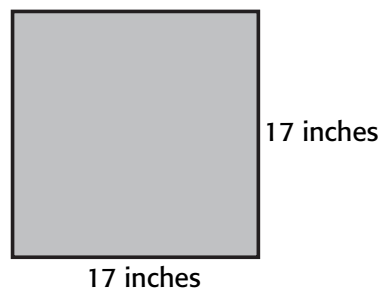
- 5** Refer to the deck in Exercise 4. Suppose Mr. Romero wants to put a railing around the perimeter of the deck. How many feet of railing will he need? (4MG1.4)
- A** 66 feet
 - B** 58 feet
 - C** 44 feet
 - D** 42 feet

- 6** Which statement about the figures is true? (4MG1.3)

Figure 1



Figure 2



- F** They both have the same area.
 - G** They both have the same perimeter.
 - H** They both have the same length.
 - J** They both have the same width.
- 7** A rectangular playground is 215 meters long and 140 meters wide. Suppose Tamika jogs around the perimeter of the playground. How far does she jog in all? (4MG1.1)
- A** 830 meters
 - B** 710 meters
 - C** 650 meters
 - D** 625 meters

Practice by Standard: Measurement and Geometry

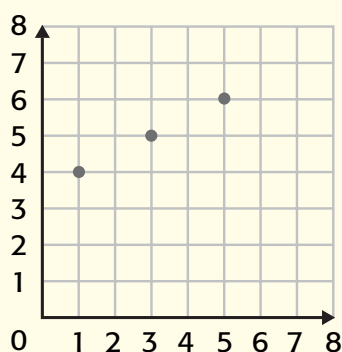
Standard Set 2.0: Students use two-dimensional coordinate grids to represent points and graph lines and simple figures.

DIRECTIONS

Choose the best answer.

QUICK Practice

- 1** Miguel plotted three points on a grid. The three points were all on the same line.



If he plots another point on the line, what could be its coordinates? (4MG2.1)

- A** (7, 6) **C** (6, 7)
B (7, 7) **D** (8, 7)

- 2** Refer to the graph in Exercise 1. If Miguel plots a point that is *not* on the line, what could be its coordinates? (4MG2.1)

- F** (9, 8) **H** (13, 11)
G (11, 9) **J** (15, 11)

QUICK Review

STRATEGY Imagine drawing a line through the 3 points to see what other points lie on the line.

Draw a line through the 3 points. Then plot each point on the coordinate grid. Which point lies on the line?

For more help with drawing points that correspond to linear relationships, see page 474.

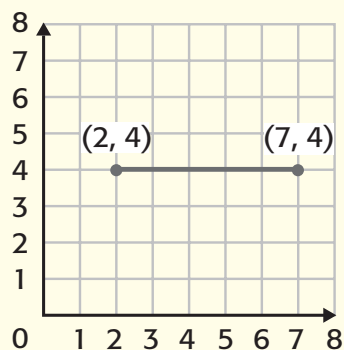
STRATEGY Imagine the line through the 3 points in Exercise 1.

Which of the points given does not lie on the line?

For more help with drawing points that correspond to linear relationships, see page 474.

QUICK Practice

- 3 Look at the line segment shown below.



What is the length of the line segment? (4MG2.2)

- A 0 units
- B 3 units
- C 5 units
- D 6 units

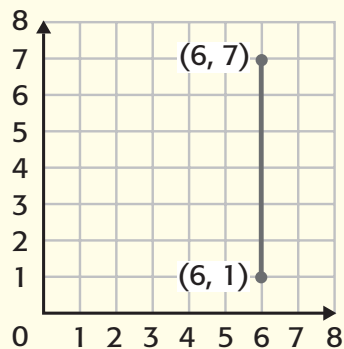
QUICK Review

STRATEGY Find the difference between the x-coordinates of the endpoints to find the segment's length.

The length of the segment can be found by simplifying the expression $7 - 2$.

For more help with finding the length of horizontal line segments, see page 474.

- 4 What is the length of the line segment shown on the grid? (4MG2.3)



- F 0 units
- G 4 units
- H 5 units
- J 6 units

STRATEGY Find the difference between the y-coordinates of the endpoints to find the segment's length.

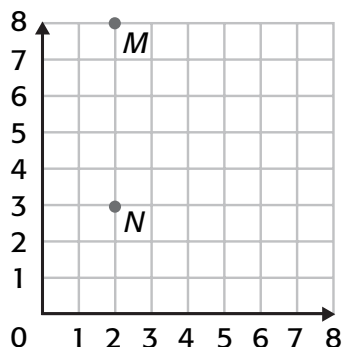
The length of the segment can be found by simplifying the expression $7 - 1$.

For more help with finding the length of vertical line segments, see page 474.

Practice on Your Own

Standard Set 2.0

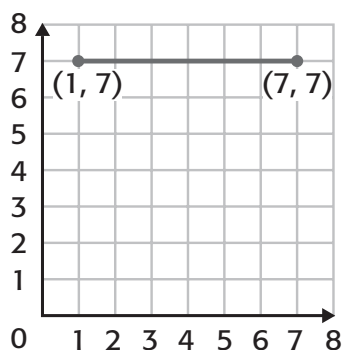
- 5** Look at the graph. Point M is at $(2, 8)$. Point N is at $(2, 3)$.



How can you find the number of units from point M to point N ? **(4MG2.3)**

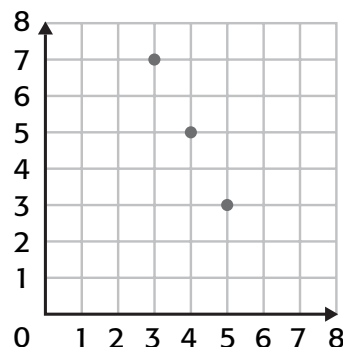
- A** Add: $8 + 3$
- B** Add: $8 + 2$
- C** Subtract: $8 - 3$
- D** Subtract: $8 - 2$

- 6** What is the length of the line segment shown on the grid? **(4MG2.2)**



- F** 6 units
- G** 5 units
- H** 4 units
- J** 3 units

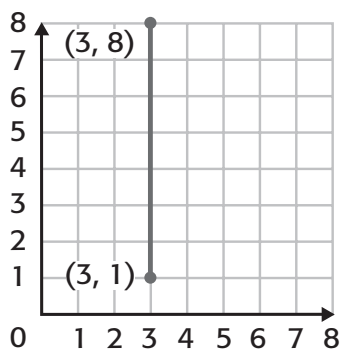
- 7** Three points that lie on the same line are plotted on the grid.



Suppose another point on the line is plotted. What could its coordinates be? **(4MG2.1)**

- A** $(6, 1)$
- B** $(7, 1)$
- C** $(6, 2)$
- D** $(7, 2)$

- 8** What is the length of the line segment shown on the grid? **(4MG2.3)**



- F** 4 units
- G** 5 units
- H** 6 units
- J** 7 units

Practice by Standard: Measurement and Geometry

Standard Set 3.0: Students demonstrate an understanding of plane and solid geometric objects and use this knowledge to show relationships and solve problems.

DIRECTIONS

Choose the best answer.

QUICK Practice

- 1 Which figure(s) below show pairs of lines that appear to be perpendicular? (4MG3.1)

Figure 1

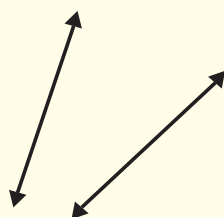


Figure 2

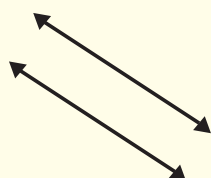
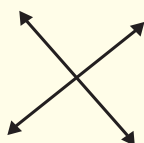


Figure 3

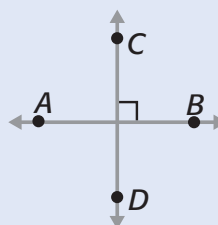


- A Figure 2 only
- B Figure 3 only
- C Figure 1 and Figure 2
- D Figure 1 and Figure 3

QUICK Review

READING HINT Perpendicular lines are lines that intersect to form right angles.

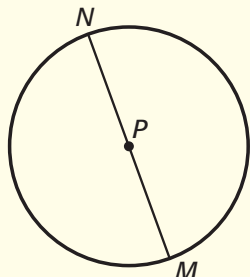
Which lines appear to intersect to form right angles?



For more help with identifying perpendicular lines, see page 392.

QUICK Practice

- 2 Look at the circle with center P .



The line segment PM appears to be (4MG3.2)

- F an arc.
- G a perimeter.
- H a radius.
- J a diameter.

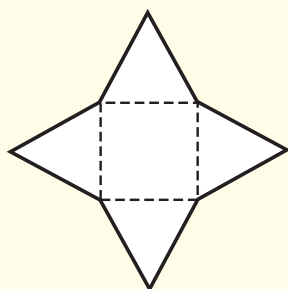
QUICK Review

READING HINT A radius of a circle connects its center to a point on the circle. A diameter passes through the center and connects two points on the circle.

What do you call a segment that connects the center of a circle to a point on the circle?

For more help with identifying the radius of a circle, see page 412.

- 3 What figure would be formed if you folded along the dotted lines? (4MG3.6)



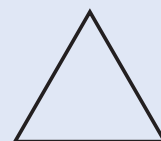
- A cone
- B cylinder
- C prism
- D pyramid

STRATEGY Imagine cutting the figure out of the paper and folding along the dotted lines. What would the folded figure look like?

The figure would have a square base and 4 triangular sides that meet at a vertex. What is the name of this three-dimensional figure?



square



triangle

For more help with identifying nets of three-dimensional figures, see page 383.

Practice on Your Own

Standard Set 3.0

- 4** What kind of triangle has three sides of different lengths? (4MG3.7)

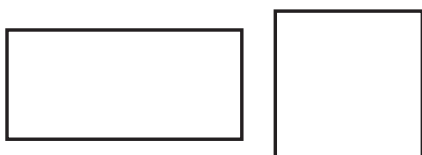
F equilateral
G isosceles
H right
J scalene

- 5** Which figures below appear to be congruent? (4MG3.3)

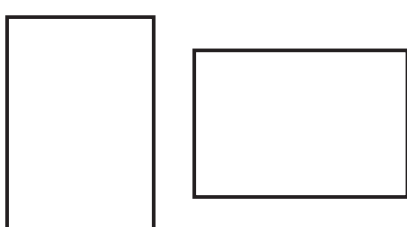
A



B



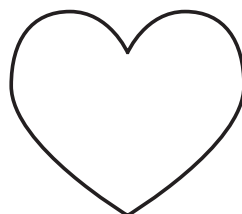
C



D



- 6** How many lines of symmetry does the figure below have? (4MG3.4)



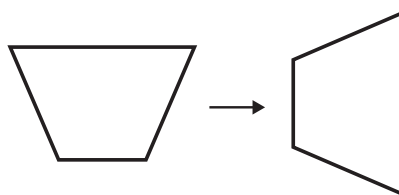
F 0

H 2

G 1

J 3

- 7** How many degrees are there in a quarter turn clockwise? (4MG3.5)



A 90°

C 270°

B 180°

D 360°

- 8** Which quadrilateral has 2 pairs of parallel opposite sides and 4 right angles? (4MG3.8)

F parallelogram
G rectangle
H square
J trapezoid

Practice by Standard: Data Analysis and Probability

Standard Set 1.0: Students organize, represent, and interpret numerical and categorical data and clearly communicate their findings.

DIRECTIONS

Choose the best answer.

QUICK Practice

- 1** A survey found that 7 people were in favor of building a new highway, 5 were opposed, and 3 were not sure. Which tally chart shows the correct results of the survey? (4PS1.1)

A

Survey Results	
In favor	
Opposed	
Not sure	

B

Survey Results	
In favor	
Opposed	
Not sure	

C

Survey Results	
In favor	
Opposed	
Not sure	

D

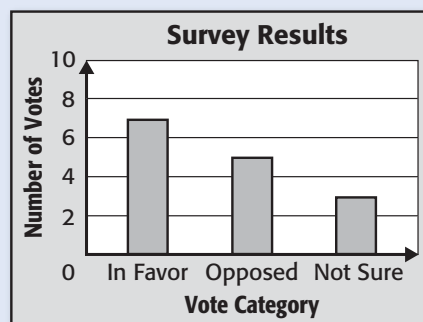
Survey Results	
In favor	
Opposed	
Not sure	

QUICK Review

STRATEGY Eliminate answer choices that do not show the correct tallies.

Which chart has 7 tally marks for in favor, 5 tally marks for opposed, and 3 tally marks for not sure?

Recall that data displayed in a tally chart can also be displayed in a bar graph.



For more help with recording the outcomes of an experiment systematically, see page 127.

QUICK Practice

- 2 The ages of the players on Gloria's volleyball team are shown below. What is the mode of the ages?

(4PS1.2)

8, 9, 8, 9, 9, 9, 10, 11, 10, 9, 9, 8

- F** 8 **H** 10
G 9 **J** 11

QUICK Review

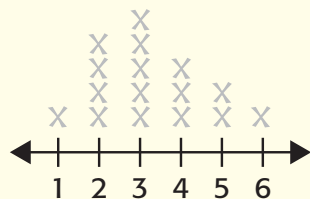
READING HINT The *mode* of a data set is the number that occurs most frequently.

Which age is most common among the volleyball players?

For more help with finding the mode of a data set, see page 130.

- 3 Mr. Baker asked the students in his homeroom how many books they read over the summer. The results are shown in the line plot below.

Number of Books



How many students read 3 books last summer? (4PS1.3)

- A** 2 **C** 4
B 3 **D** 5

STRATEGY Find the column that represents reading 3 books and count the Xs.

How many Xs are there above the number 3 in the line plot?

For more help with interpreting data displays, see page 136.

- 4 Refer to the line plot in Exercise 3. Which number of books read appears to be an outlier? (4PS1.2)

- F** 1 book
G 5 books
H 6 books
J Not here

READING HINT An *outlier* is a data point that is significantly higher or lower than the rest of the data.

Does there appear to be a data point that is significantly higher or lower than the rest of the data?

For more help with identifying outliers, see page 130.

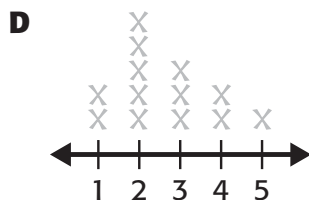
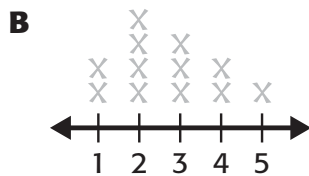
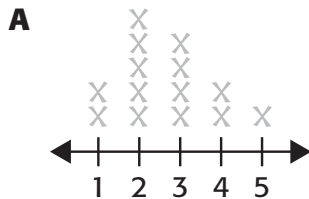
Practice on Your Own

Standard Set 1.0

- 5** The tally chart shows how many school subjects 14 students have to do homework for tonight.

Subjects	Tally
1	
2	
3	
4	
5	

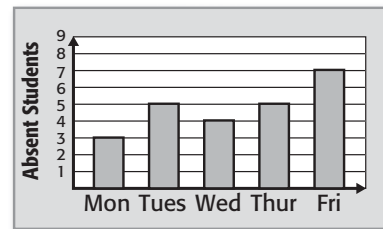
Which graph shows these results?
(4PS1.1)



- 6** Refer to the tally chart in Exercise 5. What is the mode number of school subjects that the students have to do for homework for tonight? (4PS1.2)

- F** 1 subject
G 2 subjects
H 3 subjects
J 4 subjects

- 7** Refer to the bar graph below. It shows the number of absent students each day last week at Beechwood School.



How many students were absent on Wednesday of last week? (4PS1.3)

- A** 3
B 4
C 5
D 6

- 8** Refer to the bar graph in Exercise 7. What was the median number of absent students for the week? (4PS1.2)

- F** 3
G 4
H 5
J 6

Practice by Standard: Data Analysis and Probability

Standard Set 2.0: Students make predictions for simple probability situations.

DIRECTIONS

Choose the best answer.

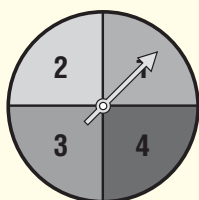
QUICK Practice

- 1** Ben has a jar with 7 green chips, 5 red chips, 4 yellow chips, and 9 blue chips in it. If he pulls out 1 chip without looking, what color is he most likely to choose?

(4PS2.2)

- A** blue
- B** green
- C** red
- D** yellow

- 2** Emma flips a coin and spins the spinner below.



How many possible outcomes are there? (4PS2.1)

- F** 2
- G** 4
- H** 8
- J** 12

QUICK Review

STRATEGY The color that has the most chips in the bag is most likely to be chosen.

What color is most common among the chips in the bag?

For more help with expressing the outcomes of probability situations verbally, see page 629.

READING HINT An *outcome* is a possible result of an experiment. For example, flipping heads and spinning a 3 is one outcome.

Draw a tree diagram to count all of the outcomes.

For more help with representing all possible outcomes of an experiment, see page 650.

QUICK Practice

- 3 The table shows the number of coins in a piggy bank.

Coin	Number
Penny	7
Nickel	4
Dime	4
Quarter	5

Suppose Brock selects a coin from the bank without looking. What is the probability that it will be a quarter? (4PS2.2)

- A $\frac{1}{4}$ C $\frac{3}{8}$
 B $\frac{3}{10}$ D $\frac{1}{2}$

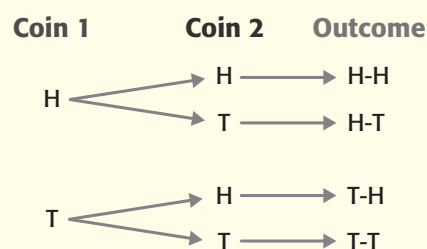
QUICK Review

STRATEGY Divide the number of favorable outcomes by the total number of outcomes.

How many coins are there in the piggy bank? How many of them are quarters?

For more help with finding the probability of a simple event, see page 634.

- 4 The tree diagram shows the possible outcomes when two coins are flipped.



How many possible outcomes are there? (4PS2.1)

- F 8 H 5
 G 6 J 4

STRATEGY Use the rightmost column of the tree diagram to count the outcomes.

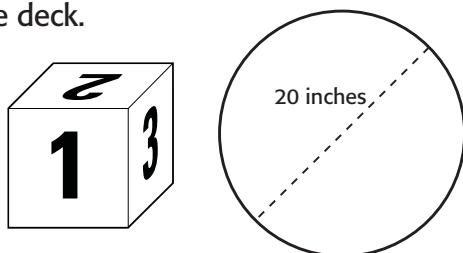
The outcomes are shown in the rightmost column.

For more help with using tree diagrams to count outcomes, see page 650.

Practice on Your Own

Standard Set 2.0

- 5** To play a board game, each player rolls a number cube and chooses a card at random from a deck. There are red, green, yellow, and blue cards in the deck.



How many possible outcomes are there on each turn? (4PS2.1)

- A** 12 **C** 24
B 16 **D** 48
- 6** Refer to the board game in Exercise 5. What is the probability that Sonia will roll a 3 and select a yellow card on her next turn? (4PS2.2)
- F** 1 out of 8
G 1 out of 12
H 1 out of 16
J 1 out of 24
- 7** David will pick one pair of pants and one sweater to wear at random from the following choices?

Pants	Sweater
Tan	Red
Black	Blue
	Orange

How many different possible outcomes are there? (4PS2.1)

- A** 4 **C** 6
B 5 **D** 8

- 8** Each week at a summer camp, so many campers are selected to be team leaders. The table shows the number of campers and team leaders during the past 3 weeks.

Team Leaders	
Campers	Leaders
120	12
150	15
190	19

Suppose there will be 180 campers at the summer camp next week. Which is the most reasonable number of team leaders to expect? (4PS2.2)

- F** 14
G 15
H 16
J 18
- 9** A pizzeria offers 3 different kinds of crust: thin, pan, and hand-tossed. Customers can also choose from among 6 different toppings: pepperoni, sausage, green peppers, mushrooms, ground beef, and olives. Suppose Tonio wants to order a 1-topping pizza. How many different possible outcomes are there?

(4PS2.1)

- A** 18
B 15
C 12
D 9

Practice by Standard: Mathematical Reasoning

Standard Set 1.0: Students make decisions about how to approach problems.

DIRECTIONS

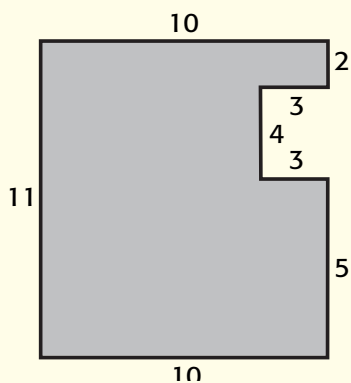
Choose the best answer.

QUICK Practice

- 1** Drew bought 5 packages of fruit juice for a soccer game. There are 4 cans of juice in each package. They sell for \$1.89 per package. How much did he spend on fruit juice altogether? (4MR1.1)

A \$10.20 **C** \$7.56
B \$9.45 **D** \$7.22

- 2** What is the area of the figure below? (4MR1.2)



F 110 square units
G 106 square units
H 102 square units
J 98 square units

QUICK Review

STRATEGY Only use the relevant information from the statement to solve the problem.

Does it matter how many cans of juice come in each package?

For more help with distinguishing relevant from irrelevant information, see page 102.

STRATEGY Break the problem into simpler parts to make solving it easier.

How can you divide the figure into parts that are all rectangles?

For more help with breaking a problem into simpler parts, see page 442.

Practice on Your Own

- 3** Maggie collects stickers. She keeps them in a picture book with the same number of stickers on each page. The table shows the total number of stickers on different numbers of pages.

Sticker Collection	
Number of Pages	Number of Stickers
1	16
2	32
3	48
4	64
5	80
6	96
7	??
8	128

How many stickers are there on 7 pages of the book? (4MR1.1)

- A** 108 stickers
B 112 stickers
C 114 stickers
D 118 stickers

- 4** Refer to Maggie's sticker collection in Exercise 3. Suppose there are 8 columns of stickers on every 2 pages of the book. Each column has the same number of stickers. How many rows of stickers are there on each page? (4MR1.2)

- F** 4
G 8
H 12
J 16

Standard Set 1.0

- 5** Look at the pattern of figures below.

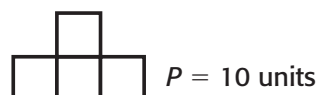


Figure 1



Figure 2



Figure 3

If the pattern is continued, what will the perimeter of the sixth figure be? (4MR1.1)

- A** 26 units **C** 30 units
B 28 units **D** 32 units

- 6** Ling ran 240 meters during recess. He weighs 120 pounds and burns 24 calories per minute while running. It took him 30 seconds to run this distance. What was Ling's average speed in meters per second? (4MR1.1)

- F** 2 meters per second
G 5 meters per second
H 8 meters per second
J 10 meters per second

Practice by Standard: Mathematical Reasoning

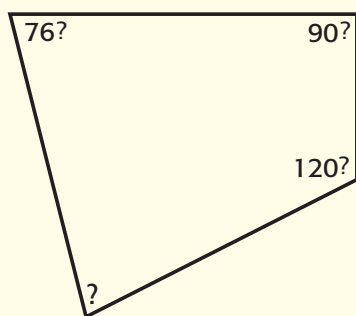
Standard Set 2.0: Students use strategies, skills, and concepts in finding solutions.

DIRECTIONS

Choose the best answer.

QUICK Practice

- 1** The sum of the angles of a quadrilateral is 360° . What is the measure of the missing angle below? (4MR2.6)



- A** 74° **C** 64°
B 71° **D** 61°

- 2** The 4th graders need to raise at least \$235 next week in the school fundraiser to win first place. Which of the following shows the amounts they can raise $\square$, to win first place? (4MR2.3)

- F** amount > 235
G amount < 235
H amount $= 235$
J amount $= 1 + 235$

QUICK Review

STRATEGY Subtract the sum of the known angle measures from 360 to find the measure of the missing angle.

This problem can be solved by simplifying the expression $360 - (76 + 90 + 120)$.

For more help with making precise calculations, see page 62.

READING HINT The phrase *at least* means greater than or equal to.

Will the 4th graders win first place if they raise \$235? Will they win first place if they raise more than \$235?

For more help with using numbers and symbols to represent mathematical reasoning, see page 28.

Practice on Your Own

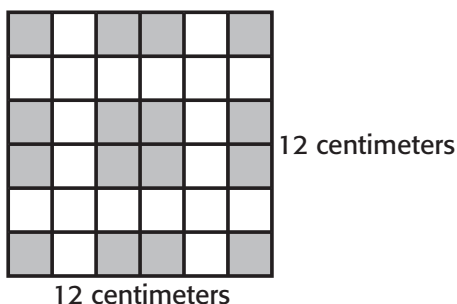
Standard Set 2.0

- 3** The table shows the costs of 4 items Timothy is purchasing at the grocery store. Estimate the total cost of these 4 items. (4MR2.1)

Grocery Shopping	
Item	Cost
Apples	\$3.05
Bread	\$2.09
Milk	\$2.85
Eggs	\$1.89

- A** \$8 **C** \$12
B \$10 **D** \$15

- 4** What is the area of the shaded part of the figure? (4MR2.2)



- F** 72 square centimeters
G 68 square centimeters
H 64 square centimeters
J 60 square centimeters

- 5** Dana has 12 dolls. Each doll has four different dresses. How could you find the total number of dresses? (4MR2.4)
- A** Add 4 and 12.
B Subtract 4 from 12.
C Multiply 4 by 12.
D Divide 12 by 4.

- 6** There are 180 players on 15 basketball teams at a tournament. Each team has the same number of players. How could you find how many players there are on each team? (4MR2.4)

- F** Add 180 and 15.
G Subtract 15 from 180.
H Multiply 15 by 180.
J Divide 180 by 15.

- 7** The table shows the total number of people who rode a certain bus each day last week.

Bus Riders	
Day	Passengers
Monday	307
Tuesday	388
Wednesday	321
Thursday	286
Friday	211

About how many people rode the bus in all last week? Round your answer to the nearest 100. (4MR2.5)

- A** 1400 **C** 1600
B 1500 **D** 1700

- 8** The average cat sleeps 18 hours a day. How could you find the number of hours in one day the average cat is awake? (4MR2.4)

- F** Add 18 and 24.
G Subtract 18 from 24.
H Multiply 18 by 24.
J Divide 24 by 18.

Practice by Standard: Mathematical Reasoning

Standard Set 3.0: Students move beyond a particular problem by generalizing to other situations.

DIRECTIONS

Choose the best answer.

QUICK Practice

- 1** José wrote the first few rows of a pattern called Pascal's Triangle. Each row has a 1 on each end. Each number is found by adding the two numbers directly above it in the previous row.

$$\begin{array}{ccccccc} & & & & 1 & & & & \\ & & & 1 & 2 & 1 & & & \\ & & 1 & 3 & 3 & 1 & & & \\ 1 & 4 & 6 & 4 & 1 & & & & \end{array}$$

What is the next row in the pattern? (4MR3.2)

- A** 1 5 10 5 1
B 1 5 10 10 5 1
C 1 6 12 15 12 6 1
D 1 6 12 12 6 1
- 2** There are 53 football players traveling to an away game. The vans hold 12 people each. Coach Brown needs to find how many vans are needed altogether. How many vans will be needed? (4MR3.1)
- F** 3 vans **H** 5 vans
G 4 vans **J** 6 vans

QUICK Review

STRATEGY Look for a pattern and predict the next number.

What happens to the numbers between the 1s with each term?

For more help with extending patterns, see page 392.

STRATEGY Eliminate answer choices that are not reasonable solutions.

What is the minimum number of vans that can hold 53 people?

For more help with evaluating the reasonableness of a solution, see page 266.

Practice on Your Own

Standard Set 3.0

- 3** In the egg toss game at the school field day, many teams are eliminated after each round.

Number of Rounds	Players Remaining
1	256
2	128
3	64
4	32
5	16
6	8

How many rounds will be needed in order to determine the winning team? (4MR3.3)

- A** 10 **C** 8
B 9 **D** 7

- 4** Coach Mata keeps track of her team's shooting statistics during the basketball season.

Shooting Statistics	
Shots Taken	Shots Made
59	20
50	17
48	16

If the team takes 75 shots in their next game, which is the most reasonable estimate for how many shots they will make? (4MR3.1)

- F** 30
G 25
H 20
J 15

- 5** Cora uses the method shown below to find the areas of triangles with different heights and base lengths.

Height: 6, Base: 4

$$\text{Area: } \frac{1}{2} \times 6 \times 4 = 12$$

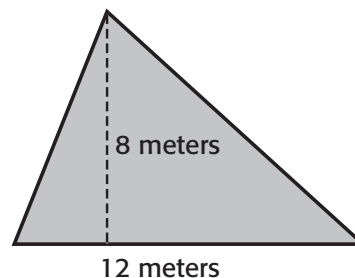
Height: 12, Base: 10

$$\text{Area: } \frac{1}{2} \times 12 \times 10 = 60$$

Height: 8, Base: 4

$$\text{Area: } \frac{1}{2} \times 8 \times 4 = 16$$

What is the area of a triangle with a height of 8 meters and a base length of 12 meters? (4MR3.2)



- A** 96 square meters
B 84 square meters
C 54 square meters
D 48 square meters

- 6** Jennie bought 5 gel pens priced at 89¢ each. Which would give the most reasonable estimate for the total cost of the 5 gel pens? (4MR3.1)

- F** 80¢ × 5 **H** 90¢ × 5
G 85¢ × 5 **J** 95¢ × 5

Practice on Your Own

Standard Set 3.0

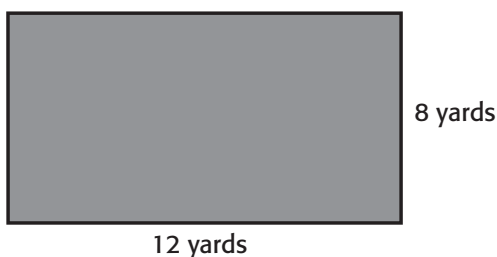
- 7** Lin created a table showing the length, width, and area of different rectangles.

Length (units)	Width (units)	Area (square units)
2	8	16
5	3	15
6	4	24
3	9	27
8	5	40

Which of the following best describes the method Lin uses to find the area of a rectangle? **(4MR3.2)**

- A** Multiply the length by the width.
- B** Divide the length by the width.
- C** Add the length to the width.
- D** Subtract the length from the width.

- 8** Use Max's method from Exercise 6 to find the area of the rectangle below. **(3MR3.2)**



- F** 20 square yards
- G** 72 square yards
- H** 84 square yards
- J** 96 square yards

- 9** There are 237 students going on a field trip. The students will travel on buses that hold 48 people each. Mr. King needs to find how many buses are needed altogether. Which of the following is the most reasonable number of buses that will be needed? **(4MR3.1)**

- A** 6 buses
- B** 5 buses
- C** 4 buses
- D** 3 buses

- 10** Vanesa has a tablecloth with an area of 48 square feet. She wants to fold the tablecloth to store it in a closet. The table below shows the area of the tablecloth after each time she folds it in half.

Number of Folds	Area (square feet)
1	48
2	24
3	12
4	6
5	3
6	■

It takes Vanesa 6 folds to get the tablecloth small enough to fit on the closet shelf. What is the area after 6 folds? **(4MR3.3)**

- F** 2.5 square feet
- G** 2 square feet
- H** 1.5 square feet
- J** 1 square foot